Copernicus Marine Service

QUALITY INFORMATION DOCUMENT

Sea Surface Temperature Production Centre

ODYSSEA Level 3S and L4 multi-year Sea Surface Temperature products for the European Northwest Shelf/ Iberia, Biscay and Irish Sea area, and Level 3S for the Global Ocean

SST_ATL_SST_L4_REP_OBSERVATIONS_010_026
SST_ATL_PHY_L3S_MY_010_038
SST_GLO_PHY_L3S_MY_010_039

Issue: 1.0

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CHANGE RECORD

When the quality of the products changes, the QuID is updated and a row is added to this table. The third column specifies which sections or sub-sections have been updated. The fourth column should mention the version of the product to which the change applies.

Issue	Date	§	Description of Change	Author	Validated By
1.0	2025/05/31	All	Creation of the document. Automatic temporal extension to one month before real time- Split of product 10_044	E. Autret, J.F. Piollé	A. Pisano

ACRONYM TABLE

Acronym	Description
CMEMS	Copernicus Marine Environment Monitoring Service
EAN	Estimated Accuracy Number
GHRSST	Group for High Resolution SST
HR	High spatial Resolution
INS	Copernicus Marine Service In-Situ
MDS	Copernicus Marine Data Store
MY	Multi-year
NRT	Near Real Time
PUM	Product User Manual
QC	quality control
REP	Reprocessing
SST	Sea Surface Temperature
TAC	Copernicus Marine Service Thematic Assembly Centre
UHR	Ultra High spatial Resolution
L3C	Level 3 (gridded) mono-satellite data
L3S	Level 3 (gridded) multi-satellite data
L4	Level 4 satellite product (gridded and gap-free data)
CCI	Climate Change Initiave
C3S	Copernicus Climate Change Service
ESA	European Space Agency

KEY DEFINITIONS

Foundation SST	Foundation Sea Surface Temperature (Foundation SST) refers to the temperature of the uppermost layer of the ocean (typically at a depth of a few centimetres to several meters) that is not affected by diurnal warming or surface cooling due to solar radiation or wind mixing. It represents the baseline or background sea surface temperature, usually estimated during the night-time or under well-mixed conditions, making it more stable and suitable for climate studies and model assimilation.
adjusted_sea_surface_temperature	daily (nighttime), satellite multi-sensor composite of foundation sea surface temperature (SST) in L3S products.
analysed_sst	Gap-free daily foundation sea surface temperature (SST) in L4 products, interpolated from L3S SSTs.

RELEVANT DOCUMENT LIST

GHRSST GDS	The Recommended GHRSST Data Specification (GDS) Revision 2.1, June 2022, https://ghrsst.org https://www.ghrsst.org/ghrsst-news/updated-ghrsst-data-specification-2-1-now-available-for-download/
Product User Manual (PUM) for ODYSSEA MY SST	Related Product User Manual, CMEMS-SST-PUM-010-026-038-029

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I EXECUTIVE SUMMARY

I.1 Products covered by this document

This document describes the quality of three Sea Surface Temperature (SST) products from the Copernicus Marine Service's SST Thematic Assembly Center (TAC) (Table 1).

The regional data are delivered as reprocessed Level 3 (L3S) and Level 4 (L4) Sea Surface Temperature (SST) products, whilst the global data are delivered as reprocessed L3S SST products.

Table 1. Copernicus Marine Service – The ODYSSEA Global Ocean and European Northwest Shelf / Iberia, Biscay & Irish Sea area (called ATL region) SST products.

Product Name	Level	Production Region
SST_ATL_PHY_L3S_MY_010_038	L3S	European North West Shelf/Iberia, Biscay & Irish Sea
SST_ATL_SST_L4_REP_OBSERVATIONS_010_026	L4	
SST_GLO_SST_L3S_MY_010_039	L3S	Global Ocean

I.2 Summary of the results

The Copernicus Marine Service ODYSSEA reprocessed L3S products provide both regional and global multi-sensor composites of sea surface temperature (SST), derived from multiple infrared instruments aboard polar-orbiting satellites. The L4 products are built from the L3S products in order to provide gap-free SST fields. The products provide an estimate of the foundation SST, which is the SST free of diurnal variability, created by the Ifremer ODYSSEA system using re-processed dataset from the European Space Agency's (ESA) SST Climate Change Initiative (CCI) and the Copernicus Climate Change Service (C3S) programe [R.2].

The products are available from 1st January 1982 to one month before real time (M-1) on a regular grid, spanning from 21° W to 13° E and 9° N to 62° N, at 0.05° resolution for the ATL area and 0.1° for Global Ocean. The ATL area is shown in Figure 1.

The products SST_ATL_PHY_L3S_MY_010_038 and SST_GLO_PHY_L3S_MY_010_039 were compared to in situ data to assess the quality of the following variable:

- **adjusted_sea_surface_temperature** – daily (nighttime), satellite multi-sensor composite of sea surface temperature (SST).

For the product SST_ATL_SST_L4_REP_OBSERVATIONS_010_026 was compared to in situ data to assess the quality of the following variable:

- **analysed_sst** – daily (nighttime), gap-free sea surface temperature (SST).

The validation of SST TAC products is exclusively based on the use of drifting buoy measurements, as recommended currently by the GHRSST (Group for High Resolution Sea Surface temperature[1]) group on satellite SST validation, and the CMEMS INSITU_GLO_TS_REP_OBSERVATIONS_013_001_b product [R.1] is the source for the drifting buoy data. The mean and standard deviation of the departures from these drifting buoy SST measurements are computed over the period 2016-2023.

[1] https://www.ghrsst.org

The **SST_ATL_PHY_L3S_MY_010_038** product is a Level3 multi-sensor sea surface temperature composite (L3S) product over the European Northwest Shelf / Iberia, Biscay & Irish Sea area at 0.05° resolution. The product is built from a collection of the Sea Surface Temperature L3C data, obtained from various sources from the ESA SST CCI and C3S dataset. This product is available from 1st January 1982 to 1 month before real time (M-1) on a regular grid (0.02° in latitude and longitude) spanning from 21° W to 13° E and 9° N to 62° N. For the period 2016 – 2023, the mean difference between the analysis and the drifting buoy reference data is -0.14 K and the standard deviation of the differences is 0.51 K.

The **SST_GLO_PHY_L3S_MY_010_039 product** is a Level-3 global multi-sensor sea surface temperature composite (L3S) product at 0.1° resolution. This monthly product is built from a collection of the Sea Surface Temperature L3C data, obtained from various sources from the ESA SST CCI and C3S satellite datasets. The product is available from 1st January 1982 to 1 month before real time (M-1) on a regular grid spanning from 180° W to 180° E and 80° S to 80°N. For the period 2016 – 2023, the mean difference between the analyses and the drifting buoy reference is -0.16 K and the standard deviation of differences is 0.51 K.

The **SST_ATL_SST_L4_REP_OBSERVATIONS_010_026** reprocessed Level 4 Sea Surface Temperatures) product is a daily, near real time, spatially complete (level 4) foundation sea surface temperature (SST) dataset over European Northwest Shelf / Iberia, Biscay & Irish Sea area at 0.05° resolution. This product is available from 1st January 1982 to 1 month before current time (M-1) on a regular grid spanning from 21° W to 13° E and 9° N to 62° N. For the period 2016 – 2023, the mean difference between the analyses and the drifting buoy reference is -0.10 K and the standard deviation of differences is 0.47 K.

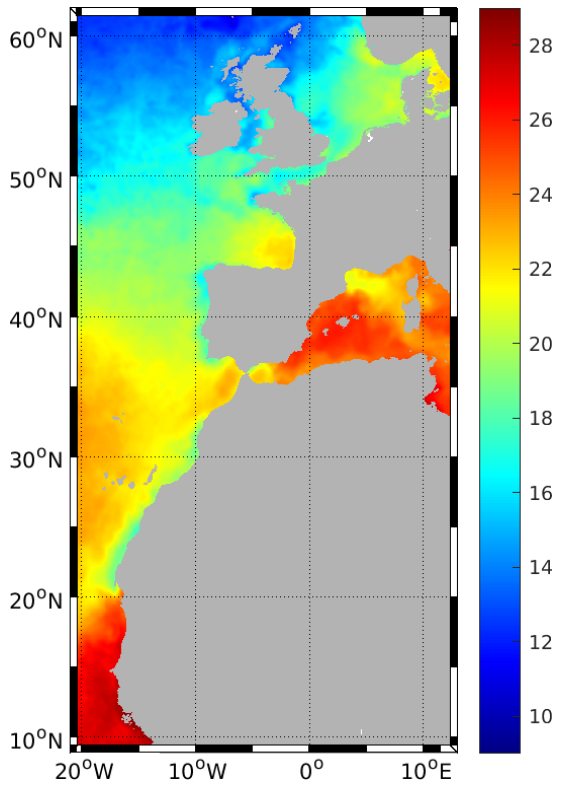

Figure 1. Area covered by the ATL product. The map shows an example of a reconstructed SST field in degree Celsius (from the SST_ATL_L4_REP_OBSERVATIONS_010_026 product, on 29/07/2014).

I.3 Estimated Accuracy Numbers

Table 2 and Table 3 below summarise the quality metrics (mean and standard deviation) for the 3 products presented in this document. The mean biases show a slight cold bias deviation, while the standard deviation values confirm that the overall error remains below or close to half a degree, which is consistent with the expected accuracy for high-quality SST products.

Table 2. Estimated accuracy numbers resulting from the validation process. Mean and standard deviation for the differences between the adjusted sea surface temperature from the L3S or the analysed SST (from the L4 reprocessed SST product) and the in-situ SST measurements provided by In-Situ TAC. Statistics were computed in the European Northwest Shelf/Iberia, Biscay & Irish Sea area for the period 17/05/2016 to 31/12/2023.

Parameter	Mean (K)	Standard deviation (K)
adjusted_sea_surface_temperature (010_038)	-0.14	0.51
analysed_sst (010_026)	-0.10	0.47

Table 3. Estimated accuracy numbers resulting from the validation process. Mean and standard deviation for the differences between the adjusted sea surface temperature from the L3S reprocessed SST product and the in-situ SST measurements provided by In-Situ TAC. Statistics were computed across the Global Ocean for the period 17/05/2016 to 31/12/2023.

Parameter	Mean (K)	Standard deviation (K)
adjusted_sea_surface_temperature (010_039)	-0.16	0.51

II PRODUCTION SYSTEM DESCRIPTION

This section summarises the characteristics of the three products addressed in this document.

II.1 Production centre details

Table 4. Details of the SST_ATL_PHY_L3S_MY_010_038 product discussed in this document.

Production centre name	**IFREMER, FR**
Product Name	SST_ATL_PHY_L3S_MY_010_038
Short Description	L3S daily maps of SST
Variables/datasets	adjusted_sea_surface_temperature
Reprocessing	yes
Geographical coverage	ATL (European Northwest Shelf/ Iberia, Biscay and Irish Sea area)
Spatial resolution	0.05 x 0.05°
Vertical coverage	surface
Temporal coverage	01/01/1982 – M-1
Temporal resolution	Daily (night-time)
Input satellites	ESA CCI & C3S products from AVHRR_NOAA satellites, ERS1/2, Envisat, METOP B, METOP C, Sentinel 3A / 3B/, AQUA-AMSR-E, GCOMW1-AMSR2
Target delivery time	monthly
Update Frequency	monthly

Table 5. Details of the SST_ATL_SST_L4_REP_OBSERVATIONS_010_026 product discussed in this document.

Production centre name	**IFREMER, FR**
Product Name	SST_ATL_SST_L4_REP_OBSERVATIONS_010_026
Short Description	L4 daily maps of SST
Variables/datasets	analysed_sst
Reprocessing	yes
Geographical coverage	ATL (European Northwest Shelf/ Iberia, Biscay and Irish Sea area)
Spatial resolution	0.05 x 0.05°
Vertical coverage	surface
Temporal coverage	01/01/1982 – M-1
Temporal resolution	Daily (night-time)
Input satellites	ESA CCI & C3S products from AVHRR_NOAA satellites, ERS1/2, Envisat, METOP B, METOP C, Sentinel 3A / 3B/, AQUA-AMSR-E, GCOMW1-AMSR2
Target delivery time	monthly
Update Frequency	monthly

Table 6. Details of the SST_GLO_PHY_L3S_MY_010_039 product discussed in this document.

Production centre name	**IFREMER, FR**
Product Name	SST_GLO_PHY_L3S_MY_010_039
Short Description	L3S daily maps of SST over global ocean
Variables/datasets	adjusted_sea_surface_temperature
Reprocessing	yes
Geographical coverage	Global ocean (GLO)
Spatial resolution	0.1x0.1 degree
Vertical coverage	surface
Temporal coverage	01/01/1982 – M-1
Temporal resolution	Daily (night-time)
Input satellites	ESA CCI & C3S products from AVHRR_NOAA satellites, ERS1/2, Envisat, METOP B, METOP C, Sentinel 3A / 3B/, AQUA-AMSR-E, GCOMW1-AMSR2
Target delivery time	monthly
Update Frequency	monthly

II.2 SST ODYSSEA-REP Processing Chain and Algorithm

This section presents the SST ODYSSEA-REP method (processing chain) used to generate the L3S and L4 SST time series. This method has been developed for the reprocessing of SST time series and is similarly applied for both regional areas (European Northwest Shelf / Iberia, Biscay & Irish Sea area) and the global ocean, differentiated by the grid resolution, 0.02x0.02 degree and 0.1x0.1 degree respectively. The following sections describes the input data, the methodological approach and some results.

II.2.1 Input data

The satellite derived SST data which are used to reconstruct the daily SST fields are extracted from the archived global ESA-CCI and C3S dataset]. As described in [R.2], this archive includes observations from:

- **Satellite thermal infra-red measurements** obtained from the Advanced Very High Resolution Radiometers (AVHRRs),
- **Along-Track Scanning Radiometers** (ATSRs),
- **Sea and Land Surface Temperature Radiometer** (SLSTR) on-board SENTINEL-3 and
- **Microwave observations**.

In the L3S product, the absence of valid SST values is mainly due to the presence of clouds. Only observations with quality level values greater than 4 (4 corresponding to acceptable and 5 corresponding to best quality measurements as described in [R.2]) have been included in the analysis. In order to avoid the use of data affected by diurnal warming, only night-time observations have been included.

II.2.2 Method

II.2.2.1 Production of daily adjusted sea surface temperature (L3S)

The L3S product is derived from a collection of sea surface temperature (SST) data acquired by various instruments (R. 2), resulting in multiple SST observations for each pixel on a given day. Due to differences in the spectral channels used and variations in instrument sensitivity, SST measurements from different sensors may exhibit slight discrepancies. These differences typically occur at a spatial scale of approximately 100 km. Directly aggregating data from these diverse sources without accounting for such biases could introduce artifacts (including discontinuities and noise) into the L3S SST products.

To ensure consistency of the daily L3S dataset, intercalibration of the various data sources is required. This process involves estimating a daily correction to be applied to SST values from each individual satellite source. First, a daily reference SST field is constructed by computing the median of all available observations. Then, for each satellite data source, a large-scale bias field is derived by calculating the difference between the observations and the daily reference, with the anomaly smoothed using a 70 km Gaussian filter. The estimated correction is subsequently subtracted from the SST values of each satellite source, producing the L3S *adjusted_sea_surface_temperature* variable.

The daily individual sensor composite files are then merged into a combined multi-sensor composite file (L3S) at the final product resolution. The merging process takes into account a hierarchy of sensors (from best to worst, again, estimated from comparisons of each source with in situ measurements). Any given cell of the final grid will contain a value from one source only, which is selected as the best. The used source is reported in the *sources_of_sst* variable of the created L3S dataset.

II.2.2.2 Production of daily analysed SST and analysis error (L4)

The reconstruction of the daily gap-free SST time series relies on the interpolation at each point of the time series with a Kalman smoother, as described in [R. 3]. This interpolation has been carried out on the time series of SST anomalies obtained by removing a first-guess SST.

The model used to generate the first-guess SST fields and the method to reconstruct the time series are presented below.

First-guess fields

The first-guess SST field used in this analysis is based on the formulation of the mean seasonal cycle and the linear long-term trend ([R. 5] and [R. 6]) at each point as shown in the following equation (Eq. 1):

$$T = A_0 + A_1 + A_2 \cos\left(2\pi/365.25(t - A_3)\right) \qquad \text{(Eq. 1)}$$

Where *T* is the sea surface temperature, *t* the number of days since 1 January 1982,

$A_0 + 365.25 * A_1/2$ can be interpreted as the mean SST at the beginning of the period, and A_1 is the slope of the linear warming. A_2 corresponds to the amplitude of the seasonal cycle, and the phase A_3 corresponds approximately to the coldest days in the year. The parameters have been estimated by minimizing the mean squared difference between the observations and the estimated temperature data.

Daily SST fields reconstruction

The reconstruction is based on the Kalman smother detailed in [R. 3], therefore this section only presents the parameters that have been estimated for the analysis.

At each point, the time series of SST anomalies (difference between SST data and first-guess SST) was considered as:

$$y_{t_1}^{t_n} = \left(y_{t_1},\ldots,y_{t_n}\right) \qquad \text{(Eq. 2)}$$

It was assumed that this new time-series (referred to the SST anomaly hereafter) is a discrete-time realization of a continuous time stationary process $Y_{t.}$ The model that was considered for $\{Y_t\}$ is introduced below. First, it was assumed that the observed SST anomaly at time t, Y_t is related to the "true" SST anomaly at time t, denoted X_t, by the measurement equation below:

$$Y_t = HX_t + \sqrt{R}\epsilon_t \qquad \text{(Eq. 3)}$$

where $\{\epsilon_t\}$ is a Gaussian white noise sequence with zero mean and unit variance. In practice R represents the variance of the observation error and H allows a transformation between the state and the observations. Then it was again assumed that the latent process $\{X_t\}$ is a simple Ornstein-Uhlenbeck process; that is, a stationary solution of the following stochastic differential equation:

$$dX_t = -\lambda X_t dt + \tau dW_t \qquad \text{(Eq. 4)}$$

Where $\{W_t\}$ denotes a standard Brownian motion. $\lambda > 0$ is the time correlation (in day) or feedback parameter which represents the slowly evolving transfer of heat and, $\tau > 0$ is the variability coming from atmospheric fluctuations. Hereafter, we denote σ, the variance of the stationary distribution:

$$\sigma = Var(X_t) = \frac{\tau^2}{2\lambda} \qquad \text{(Eq. 5)}$$

with $\{X_t\}$ a Markov process which satisfies, for $i \in \{2,\ldots,n\}$,

$$X_{t_i} = M_{\Delta_i} X_{t_{i-1}} + \sqrt{\left(Q_{\Delta_{i-1}}\right)}\eta_{t_i} \qquad \text{(Eq. 6)}$$

$$\text{with } M_{\Delta_i} = \exp(-\lambda\Delta_i), Q_{\Delta_i} = \sigma^2\left(1 - M_{\Delta_i}^2\right) \qquad \text{(Eq. 7)}$$

and $\left\{\eta_{t_i}\right\}_{i\in\{2,\ldots n\}}$ a Gaussian white noise sequence with zero mean and unit variance independent of $\left\{\epsilon_{t_i}\right\}_{i\in\{1\ldots n\}}$. Here, since the temporal sampling is irregular, the time steps $\Delta_{1,}\ldots,\Delta_n$ are not equal. $\{Y_t\}$ is a stationary Gaussian process with zero mean and covariance function:

$$Cov\left(Y_t, Y_{t'}\right) = H^2\sigma_2\exp\left(-\lambda\left|t - t'\right|\right) + R1_0\left(t - t'\right) \qquad \text{(Eq. 8)}$$

It was deduced that the distribution of the observed sequence $\left(y_{t_1},\ldots,y_{t_n}\right)$ is a multivariate Gaussian distribution with a zero mean and covariance matrix which can be expressed from the unknown parameters H, R, $\sigma^2$and λ. Here H has been fixed to 1. The covariance function corresponds to a classical model in spatial statistics, since an exponential model with nugget R, sill $\sigma^2 + R$ and range $1/\lambda$ was retrieved. In the analysis, the parameter estimation has been done by a method based on the moments ([R.3])

The reconstruction with the Kalman smoother has been performed with R, σ^2 and $\widehat{M}_1$,estimated from the dataset (where $\widehat{M}_1 = \exp\left(-\hat{\lambda}\right)$). Then, the resulting SST fields have been convolved with a Gaussian kernel of standard deviation 3.5 km.

III VALIDATION FRAMEWORK

This section introduces the validation protocols used to assess the SST products.

The validation of SST TAC products is only based on drifting buoy measurements, as recommended currently by the GHRSST[3] group on satellite SST validation (STVAL). This validation study uses drifting buoy measurements provided by the CMEMS In-situ TAC (CMEMS INSITU_GLO_TS_REP_OBSERVATIONS_013_001_b product [R. 1]). As the L3S SST products are based on satellite observations only, drifting buoy measurements can be considered as independent validation data.

The mean and standard deviation from drifting buoy SST measurements are computed over the European Northwest Shelf/Iberia, Biscay & Irish Sea area and the global ocean. The SST product minus observation differences is calculated by bi-linearly interpolating the analysis to the observation locations for that day. The mean and standard deviation of these differences are then averaged.

Only night-time measurements are kept, in order to be consistent with the data screening policy for the input satellite data.

[3] https://www.ghrsst.org/

IV Validation results

This section shows the validation results for three SST products. This section summarises the quality metrics for the products presented in this document, and presents the bias and standard deviation of the differences between satellite SST observations and drifting buoy measurements for the ATL region over the period 01/01/2016 to 31/12/2023. The production system and framework for quality evaluation are described in Section III and results are described in Table 2 (section I.3).

Please note that the number of observations delivered in the product is not homogeneous in time and space. Indeed, the availability of SST observations for a given day depends mainly on the number of instruments in operation and the cloud coverage, since the input dataset is composed of satellite measurements from infrared sensors (which are only available in cloud-free conditions).

IV.1 Atlantic area (SST_ATL_PHY_L3S_MY_010_038, SST_ATL_SST _L4_REP_OBSERVATIONS_010_026)

For these two datasets, the spatial distribution of SST observations in the product is presented in Figure 2. The figure shows better coverage in the Mediterranean Sea and along the Iberian and African coastal regions between 1982 and 2023. In the north-western part of the area, there are very few observations over the entire period, due to the presence of clouds. The present version of the product includes microwave data from 2002 to 2017 ([R.2]), improving significantly the coverage in this region compared to the previous version.

The number of matchups vary in time for both products (L3S and L4), as shown with solid grey line in Figures 3 and Figure 4 respectively where the monthly mean bias (green solid line in both figures) are not necessarily correlated to the matchup numbers. The differences between the satellite product and the SST from drifting buoys are more or less homogeneous in time according to Figure 3 and Figure 4. The figures also show the time series of mean differences over the area from 2016 to 2023.

The spatial distribution for the L3S product of the mean biases (differences averaged in 0.25° x 0.25° boxes) for the entire period is shown in Figure 5a and the spatial distribution of the number of matchups is shown in Figure 5b. The spatial distribution of the matchups is not homogeneous, with very few matchups in the Northern part of the area and in the Canary upwelling region. The matchups are mainly located in the Mediterranean Sea and in the Gulf of Biscay . The SST differences are small, with mean differences < 0.1 °C. In the Atlantic Ocean, the mean differences are small as well, but slightly negative. Those negative biases could be due to observations close to clouds or due to possibly undetected clouds. The negative differences visible in the Canary upwelling region are caused by the lack of a sufficient sample of matchups due either to undetected clouds or to observations that are not exactly at the same location[4] in a region where SST has a very high variability in space and time.

[4] satellite values are an average over an area with low variability, and drifter values are representative of a point

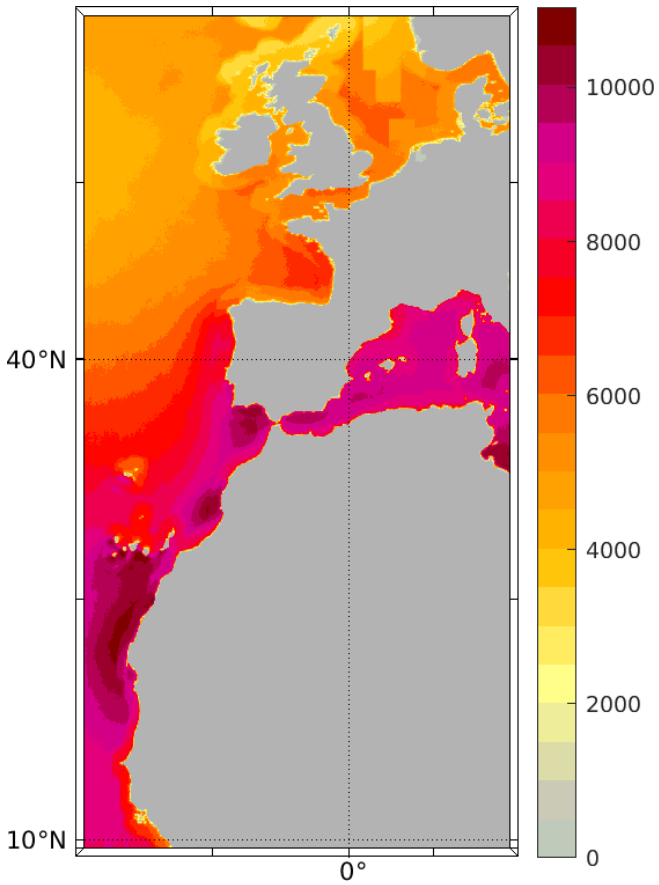

Figure 2. Spatial distribution of observations (1982-2023). The number of observations is shown in the legend.

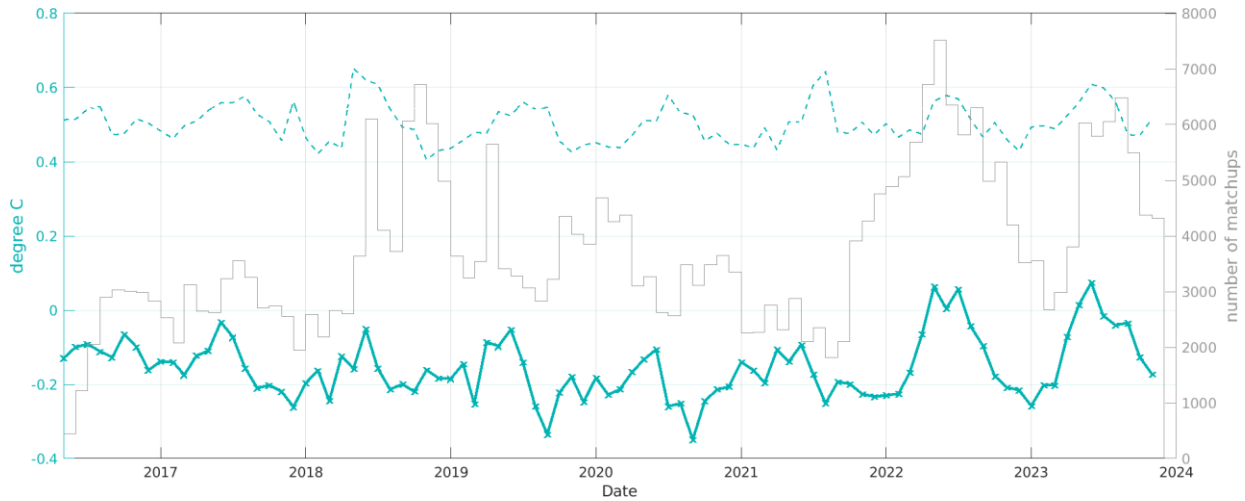

Figure 3. Time series of the monthly mean difference (solid green line) between the L3S sea surface temperature and the drifting buoys. The number of observations per month (solid grey line) and the standard deviation of the differences (dotted green line) are also shown.

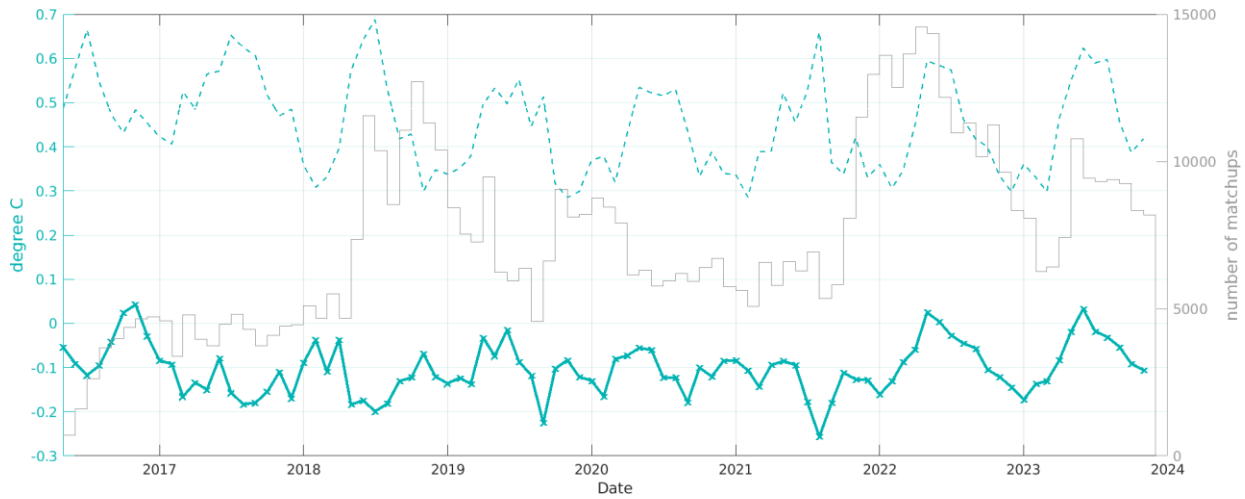

Figure 4. Time series of the monthly mean difference (solid green line) between the L4 sea surface temperature and the drifting buoys. The number of observations per month (solid grey line) and the standard deviation of the differences (dotted green line) are also shown.

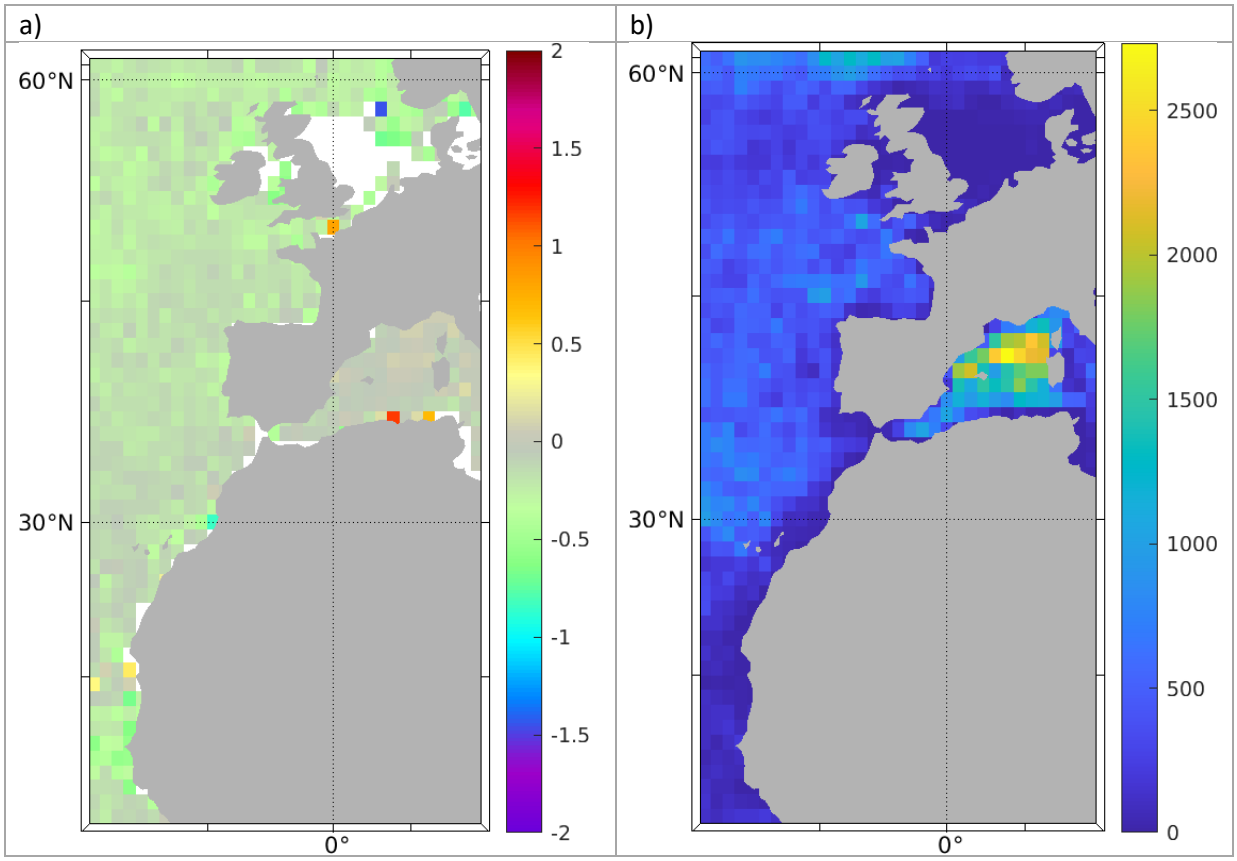

Figure 5. Spatial distribution of mean biases (differences averaged per 0.25°x0.25° boxes) in °C for the L3S product (a) and spatial distribution of the number of matchups for the L3S product (b).

IV.2 Global Ocean dataset (SST_GLO_PHY_L3S_MY_010_039)

The spatial distribution of the number of SST observations in the GLO L3S product over the 1982-2023 time period is presented in Figure 6. The figure shows that there are very few observations for the entire period in high latitude regions and equatorial regions. Missing observations are mainly due to cloudy conditions.

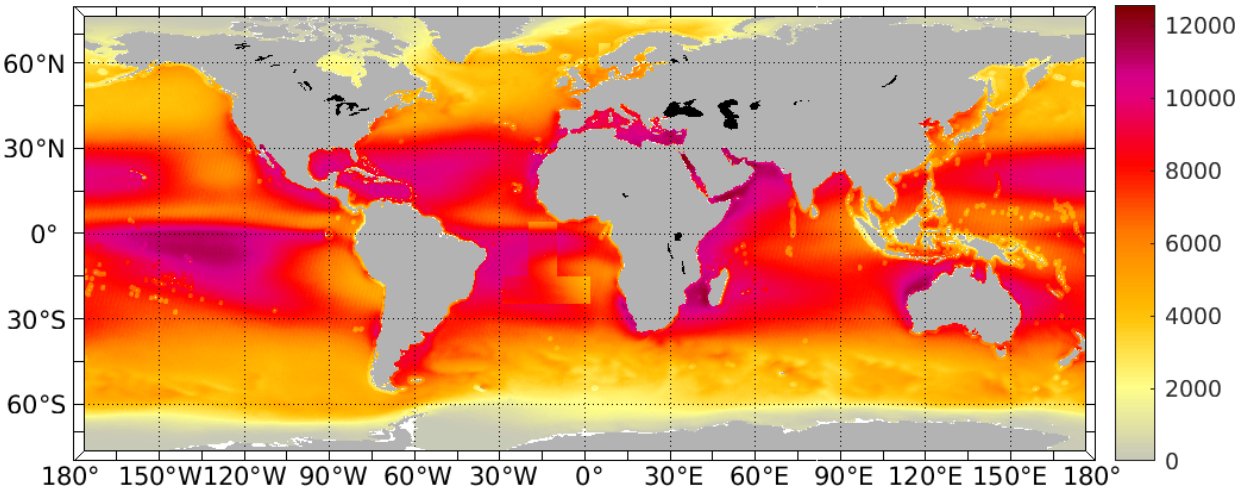

Figure 6. Spatial distribution of the number of satellite SST observations within the 1982-2023 time period.

The bias and standard deviation of the differences between satellite SST observations and drifting buoy measurements for the global region during the period 01/05/2016 to 31/12/2023 are presented in Table 3 (section I.3). The number of matchups is 5,491 541. The spatial distribution of the mean biases (differences averaged in 2.5°x2.5° boxes) for the entire period is shown in Figure 7 and the spatial distribution of the number of matchups is shown in Figure 8. It can be noted that the spatial distribution of the matchups is not homogeneous in the global Ocean. The mean difference is usually small but slightly negative (e.g., with less matchups -bright green, blue and violet colours in the Figure 7-) in high latitude areas and the Benguela upwelling region, off the coast of South Africa. In certain areas, larger negative values can be observed and could be due to observations close to clouds or possibly undetected clouds.

In the equatorial region of the African coast, the negative bias can caused by possible errors in the atmospheric corrections (high water vapor content and Saharan dust content in the atmosphere). In regions exhibiting high variability, high values of SST differences can be due to undetected clouds or to the comparisons between observations that are not exactly at the same location in a region where SST has a very high variability in space and time. The differences between the satellite product and the SST from drifting buoys are quite homogeneous in time according to Figure 9. This figure shows the time series of mean differences over the global ocean from 2016 to 2023.

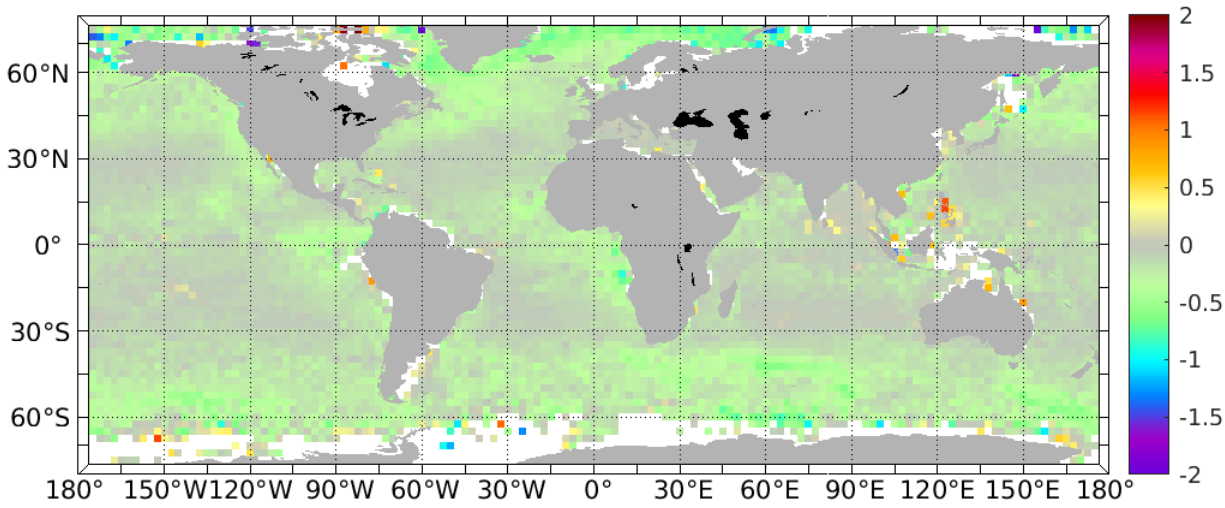

Figure 7. Map of mean SST differences (in °C) over the area from 2016 to 2023.

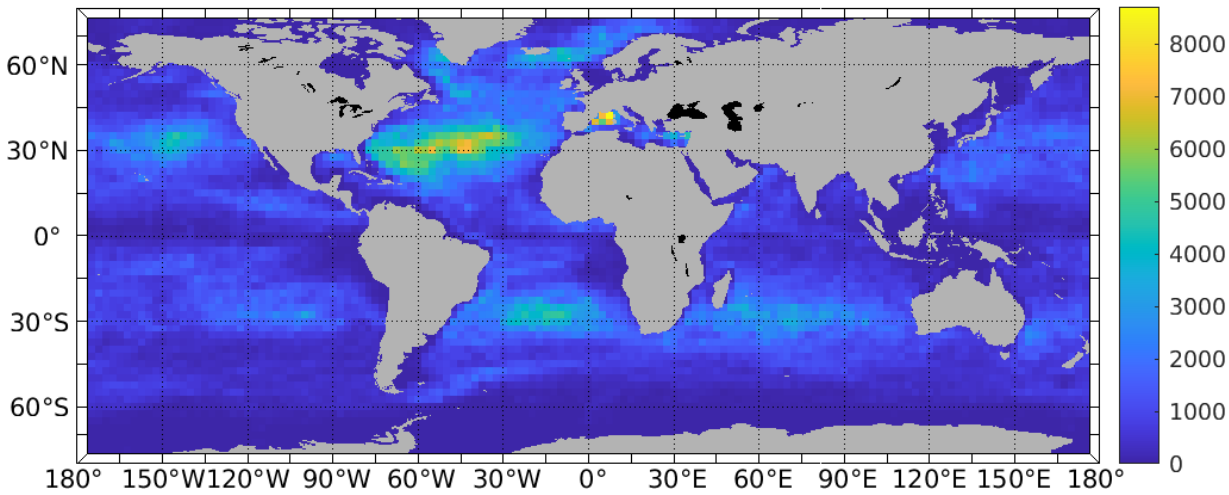

Figure 8. Spatial distribution of the number of matchups (2016-2023).

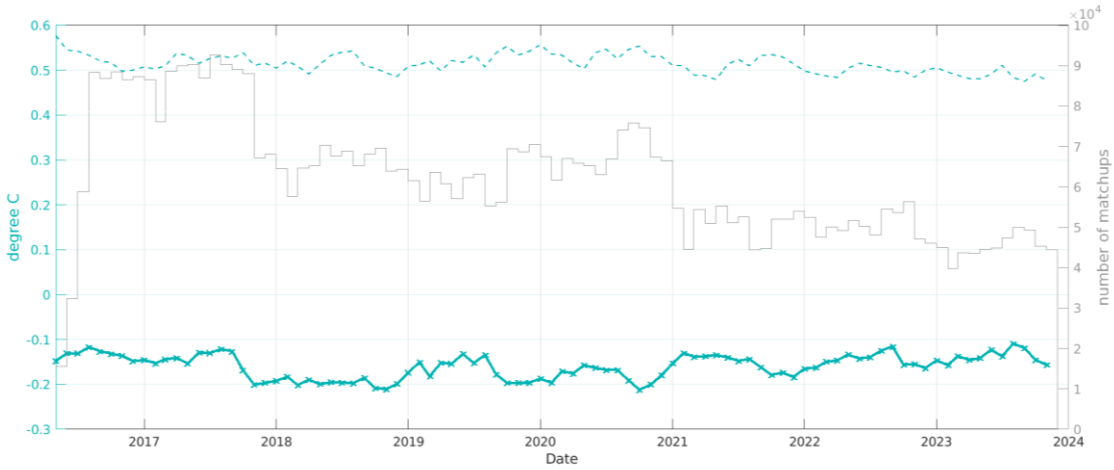

Figure 9 Time series of the monthly mean difference (solid green line) between the global L3S sea surface temperature and the drifting buoys. The number of observations per month (solid grey line) and the standard deviation of the differences (dotted green line) are also shown.

V SYSTEM'S NOTICEABLE EVENTS, OUTAGES OR CHANGES

This section is dedicated to describe and track changes that may occur in the current operational system. In includes changes due to outages (data gaps), upstream data (e.g., addition or loss of a sensor, inclusion/removal of in-situ platforms), reprocessing, corrections, assimilated data, boundary conditions, among other changes that may impact product quality (Table 7).

Table 7. Change description.

Date	Change/Event description	System version	Other
16/06/2022	New product	2.0	
16/06/2023	Temporal extension	2.0	
31/05/2024	Temporal extension (1982-2023) and upstream change (v3 CCI/C3S)	3.0	
31/05/2025	Automatic temporal extension to one month before current time	3.0	

VI QUALITY CHANGES SINCE PREVIOUS VERSION

This section aims to provide a short summary of the main changes in Product Quality between the previous and current version.

The new L3S time series dataset (i.e., times series of SST fields) includes a significantly higher number of observations compared to the previous version released in November 2023, primarily due to the incorporation of microwave data spanning from 2002 to 2017. This added data improves the sampling of mesoscale sea surface temperature (SST) features, particularly in cloudy areas. However, it does not enhance data coverage in coastal regions, as the microwave data is unavailable within a 75 km coastal band. Despite the inclusion of this new data source and the updated CCI/C3S dataset, the overall results remain very similar to those in the previous version.

VII REFERENCES

Ref.	Document Name	Document Reference	Issue	Date
[R.1]	Szekely, 2022, Product User Manual for For Global Delayed Mode In situ Dataset INSITU_GLO_TS_REP_OBSERVATIONS_013_001_b			2022
[R.2]	Embury, O., Merchant, C. J., Good, S. A., Rayner, N. A., Høyer, J. L., Atkinson, C., ... & Donlon, C. (2024). Satellite-based time-series of sea-surface temperature since 1980 for climate applications. *Scientific Data*, *11*(1), 326.			2024
[R. 3]	Tandeo, Pierre, Pierre Ailliot, and Emmanuelle Autret. "Linear Gaussian state-space model with irregular sampling: application to sea surface temperature." Stochastic Environmental Research and Risk Assessment 25.6, 2011, 793-804.			2011
[R. 4]	B. Saulquin and F. Gohin, « Mean seasonal cycle and evolution of sea surface temperature from satellite and in-situ data in the English Channel for the period 1986-2006, International Journal of Remote Sensing, Vol 31, No15, 4069-4093, 2010			2010
[R. 5]	Gohin Francis, Saulquin Bertrand, Bryere Philippe. Atlas de la Température, de la concentration en Chlorophylle et de la Turbidité de surface du plateau continental français et de ses abords de l'Ouest européen. Ifremer, 2010			2010